FOCAL: Forgery-Centric One-Class Artifact Learning for Out-of-Distribution Deepfake Detection

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Abstract

Existing deepfake detection methods learn facial representations that capture the identity, semantics, or geometric structure of real faces. We argue that such face-centric representations can introduce identity and semantic biases, encoding appearance attributes that correlate with a person's identity rather than with the image's manipulation, thereby limiting transferability to unseen forgery types. We propose **F**orgery-Centric **O**ne-**C**lass **A**rtifact **L**earning (**FOCAL**), a representation learning framework that models manipulation artifacts directly rather than generic facial semantics. FOCAL is trained exclusively on forged images: an encoder-decoder reconstructs spatial artifact maps from fake inputs, forcing the encoder to learn *where and how* manipulation has occurred. This reconstruction objective is complemented by spatial and frequency-domain contrastive losses that encourage invariance to input perturbations while preserving discriminative forgery cues. Because the encoder captures a compact forgery-artifact distribution, real faces unseen during training naturally fall outside. We exploit this property by proposing the Dynamic-Centroid Mahalanobis Distance (DCMD), which enables classifier-free, zero-shot detection without target-domain adaptation. FOCAL surpasses state-of-the-art methods on standard cross-dataset benchmarks in both detection AUC and pixel-level forgery despite fake-only training. This clearly demonstrates that forgery-centric representations yield more transferable features than approaches anchored to real-face distributions.

CCS Concepts

• **Computing methodologies** → *Image manipulation.*

Keywords

Deepfake Detection, One-Class OOD Detection

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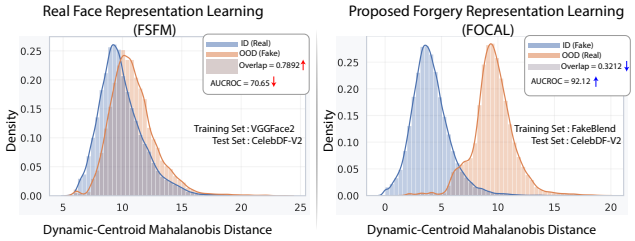


Figure 1: Comparison of representation distributions on DFDC (no fine-tuning). (a) Face-centric methods such as FSFM, trained on identity, semantics, or real-face reconstruction, show significant overlap between real and fake scores, limiting generalization. (b) FOCAL learns forgery-centric representations via fake-class artifact reconstruction, yielding clearer separation and improved generalization.

1 Introduction

Deepfake generation methods have advanced rapidly, posing severe threats to digital media integrity, individual privacy, and public trust [13, 39, 51]. Also, deepfake detection methods [2, 8–11, 23, 26, 30–33, 35, 43, 44, 47, 48, 58, 59] have made significant progress. However, a persistent challenge still remains: models trained on one forgery type or dataset often fail to generalize to unseen manipulations encountered in the wild [45, 64]. This cross-dataset generalization gap remains arguably the most critical bottleneck for real-world deployment of deepfake detectors. Recent studies have attempted to address this challenge by learning general facial representations through self-supervised pretraining [24, 28, 54, 61], assuming that a pre-trained encoder captures rich facial structure for forgery-distinguishable features and yields transferable features for downstream forgery detection. However, we argue that methods such as FSFM [61], and OC-FakeDetect [28], are fundamentally anchored to modeling *what real faces look like* during the pre-training phase, resulting in learning facial geometry, identity, and structure from authentic images rather than capturing *what forgeries look like*. This design choice limits their learned representations to facial identity features, which may not effectively discriminate across diverse

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forgery types in downstream binary classification. As illustrated in Fig. 1(a), when FSFM is trained on real images (VGGFace2) and evaluated on the unseen CelebDF-V2 dataset, the in-distribution and out-of-distribution DCMD distance distributions exhibit a high overlap of 0.7892, with an AUC-ROC of only 70.65%, indicating poor separation between real and forged samples and confirming that real-face representations alone are insufficient for generalizable forgery detection. We argue that these face-centric designs introduce two inductive biases. First, **identity bias**, where features encode appearance attributes (e.g., skin texture, shape, illumination) tied to identity rather than manipulation. Second, **semantic bias**, a representation that captures high-level cues such as expression, pose, and structure, overlooks that structure is not necessarily sensitive to the low-level, spatially localized perturbations introduced by forgery processes. Both biases degrade cross-dataset transferability, as the distributional shift between the source and target domains is predominantly driven by identity and geometric variation rather than forgery-specific cues.

This motivates a fundamental shift in deepfake detection: rather than learning representations of *faces* and capturing forgery identities in downstream fine-tuning, we propose learning representations of *forgeries* directly using forged images with forgery signals. As shown in Fig. 2, face-centric models such as the Face Security Foundation Model (FSFM) entangle forgery types, whereas **F**orgery-Centric **O**ne-**C**lass **A**rtifact **L**earning (FOCAL) learned discriminative clusters that are aligned with manipulation methods. Our hypothesis is that artifact-focused representations generalize better across datasets. Consistent with Fig. 1(b), FOCAL, trained only on fake samples from the FBI and tested on CelebDF-V2, it reduces DCMD overlap to 0.3212 (vs. 0.7892 for FSFM) and achieves 92.12% AUC-ROC, confirming that forgery-aware representations provide markedly better separation between in-distribution (fake) and out-of-distribution (real) samples. To this end, we propose **FOCAL**, a representation learning framework for deepfake detection and localization trained exclusively on forged images. Existing augmentation-based approaches, such as Self-Blending Images (SBI) [54], simulate forgeries by physically editing real faces, thereby failing to faithfully reproduce the artifacts produced by actual generative manipulation pipelines, limiting their generalizability to genuine forgery artifacts.

To address this limitation, we introduce the Fake Blending Images (FBI) dataset, which follows a procedure similar to SBI but blends a synthetically generated image of the same identity source from the Faceforensics++ (FF++) [51] real set using a randomly selected facial landmark mask. The resulting images exhibit blending artifacts that more closely resemble those produced by real forgery pipelines [54], providing a stronger training signal. An encoder-decoder architecture is then trained to reconstruct the forgery artifact region extracted from these blended images shown in Fig. 4. Crucially, the decoder's reconstruction target is itself a spatial artifact map, meaning the architecture inherently learns to localize forged regions as a byproduct of its representation-learning objective, thereby unifying detection and localization within a single framework using synthetic mask localization supervision via the proposed Artifact Reconstruction Loss (ARL). To further strengthen forgery sensitivity, we introduce two complementary contrastive objectives: (i) a spatial contrastive loss that enforces consistency between plain and augmented views of the same forgery, and (ii) a frequency-domain contrastive loss that exploits the distinctive spectral signatures left by different manipulation processes. Finally, we cast detection as an out-of-distribution (OOD) problem: since FOCAL learns a compact representation of forgery artifacts, real faces unseen during training naturally fall outside this learned distribution. We quantify this deviation using the Dynamic-Centroid Mahalanobis Distance (DCMD) [36], which enables classifier-free detection and generalizes to unseen forgery types without target-domain adaptation. FOCAL achieves a 4.6% higher average AUC than existing state-of-the-art approaches on standard cross-dataset benchmarks while also providing pixel-level forgery localization, and we present our main contributions as follows:

- We introduce FOCAL, a forgery-centric one-class representation learning framework that explicitly models manipulated regions through artifact map reconstruction, jointly learning detection and localization representations rather than relying on generic face semantics or identity-preserving embeddings.
- We formulate deepfake detection as OOD problem with classifier-free detection without downstream fine-tuning. The proposed approach achieves state-of-the-art performance on unseen datasets underpinned by a dual contrastive learning strategy that jointly optimizes augmentation-aware spatial and frequency domain objectives to produce representations invariant to perturbation and discriminative of forgery-specific spectral and spatial artifacts.
- The decoder head further provides pixel-level forgery localization as an intrinsic byproduct of the artifact reconstruction objective, thereby connecting representation learning with spatial-manipulation awareness. Extensive experiments show strong OOD generalization on deepfake detection and localization via learning forgery-aware representations.

2 Related Work

2.1 Deepfake Detection and Generalization

Early deepfake detectors relied on convolutional architectures such as XceptionNet [51] and frequency-based analysis [40, 49], achieving high in-domain accuracy but degrading significantly on out-of-distribution (OOD) data. Subsequent efforts addressed generalization through multi-scale disentanglement [37], domain adaptation [64], parameter-efficient fine-tuning of foundation models [20], and token-level augmentation [16]. Recent methods leverage pretrained vision-language models, such as CLIP [50], via forensic adapters [12] and facial component priors [20] to improve cross-dataset transferability. While these approaches demonstrate improved generalization, they primarily adapt pretrained features for binary classification without explicitly learning manipulation-specific representations, leaving them susceptible to spurious correlations between face content and forgery labels.

2.2 Face Representation Learning for Forgery Detection

A recent approach pretrains face encoders using self-supervised objectives and subsequently finetunes them for detection. FSFM [61] introduces a self-supervised facial representation foundation model that jointly learns face reconstruction and manipulation discrimination, providing a generalizable initialization for downstream

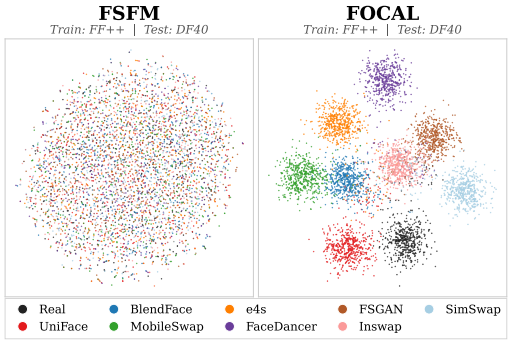


Figure 2: T-SNE plot comparison reveals improved forgery detection in DF40 manipulations using FOCAL, by training on fake-only representations using FaceForensics++ (FF++) and testing on DF40 manipulations. FSFM treats each sample as a face attribute and shows no separation after learning facial representations.

detectors. DeLoCaTe [24] integrates detection and localization by reconstructing genuine facial regions and interpreting reconstruction residuals as forgery indicators. SBI [54] generates synthetic training data by blending real faces to learn boundary artifacts. Face X-Ray [38] learns to predict blending boundaries from face composites. While these methods have advanced the field significantly, their learned representations are primarily guided by facial semantics, reconstruction, or blending structure, with forgery artifacts emerging only as indirect supervisory signals. FSFM reconstructs real face regions; DeLoCaTe [24] reconstructs authentic facial structure; SBI [54] and the Face X-Ray [38] model blending boundaries relative to real face geometry. The forgery information captured by these representations is incidental, a by-product of learning what real faces look like rather than a direct objective. Our work departs from this paradigm by making forgery reconstruction the *primary* objective in representation learning.

2.3 Artifact-Based and Localization-Aware Detection

Several methods incorporate spatial manipulation awareness. Localization detectors [55] produce pixel-level forgery masks, while multi-task frameworks jointly optimize detection and localization [25, 56], showing that spatial cues improve classification. However, localization is treated as auxiliary to classification. In contrast, FOCAL uses artifact reconstruction as the *sole* pretraining objective, learning features grounded in manipulation evidence.

2.4 One-Class and Anomaly-Inspired Detection

One-class classification [52] and anomaly detection [28] model a single (typically real) class and flag deviations as anomalies. In deepfake detection, this includes one-class SVMs [1], autoencoders modeling real-face distributions, and Mahalanobis distance-based OOD detection [36], which identifies outliers via feature-space distances but relies on fixed thresholds that limit generalization. We invert this formulation by modeling the distribution of *forgery artifacts* rather than real faces and perform detection in this representation space using the DCMD to better separate ID from OOD. This forgery-centric one-class approach, to our knowledge, has not been explored in deepfake detection and yields features that directly encode evidence of manipulation rather than assuming forgeries are merely anomalous.

3 Our FOCAL Framework

3.1 Overview and Motivation

Existing representation learning methods for deepfake detection [24, 28, 61] learn features by modeling what real faces look like through reconstruction, identity encoding, or semantic parsing and treat forgery detection as a downstream task. We argue that this real face-centric design limits generalization because the resulting features encode facial identity and semantic information that varies across datasets, rather than the manipulation-specific artifacts that are the actual target of detection. FOCAL tackles this problem by learning representations directly through modeling *what forgeries look like*. The proposed framework pretrains only on fake images from the proposed Fake Blending Images (FBI) dataset, which provides synthetic fingerprints and masks, enabling the extraction of fake regions with synthetic traces. The pipeline consists of three components: (i) a CLIP-based vision encoder ViT-L/14 [50], (ii) a masked autoencoder decoder [21], that reconstructs the forgery region through the proposed Artifact Reconstruction Loss (ARL), and (iii) augmentation-aware spatial and frequency-domain contrastive objectives. We frame classification as an OOD problem: the frozen encoder produces features that are scored against the learned forgery distribution using the DCMD distance, enabling classifier-free detection without target-domain adaptation. Fig. 3 illustrates the complete pipeline: the pretraining phase learns forgery-aware representations exclusively from fake images, while the detection phase uses the frozen encoder's features to distinguish in-distribution (fake) samples from out-of-distribution (real) samples without ever exposing the encoder to real images during training.

3.2 Problem Formulation

We begin by formally defining the training environment for our forgery detection framework. Let $\mathcal{D}_{\text{train}} = \{x_i^{\text{f}}, x_i^m\}_{i=1}^N$ denote a training set of fake facial images from a single proposed source FBI dataset, where x_i^{f} is a manipulated face and x_i^m is its corresponding mask. Our goal is to learn an encoder $f_{\text{enc}} : \mathcal{X} \to \mathbb{R}^d$ whose feature space $\mathcal{Z}$ satisfies two properties: (1) features of forged images cluster tightly in $\mathcal{Z}$, forming a compact forgery-aware distribution; and (2) features of authentic images are well-separated from this cluster. Critically, these properties should hold *across datasets*, including forgery types and identities not seen during training. We achieve this by training an encoder (f_{enc}) to produce features from which a decoder (f_{dec}) can reconstruct the spatial forgery area, while simultaneously enforcing forgery-consistent structure via contrastive learning. Unlike prior face-centric methods that train on both real and fake classes, FOCAL trains *only on fake images*, thereby enabling one-class representation learning.

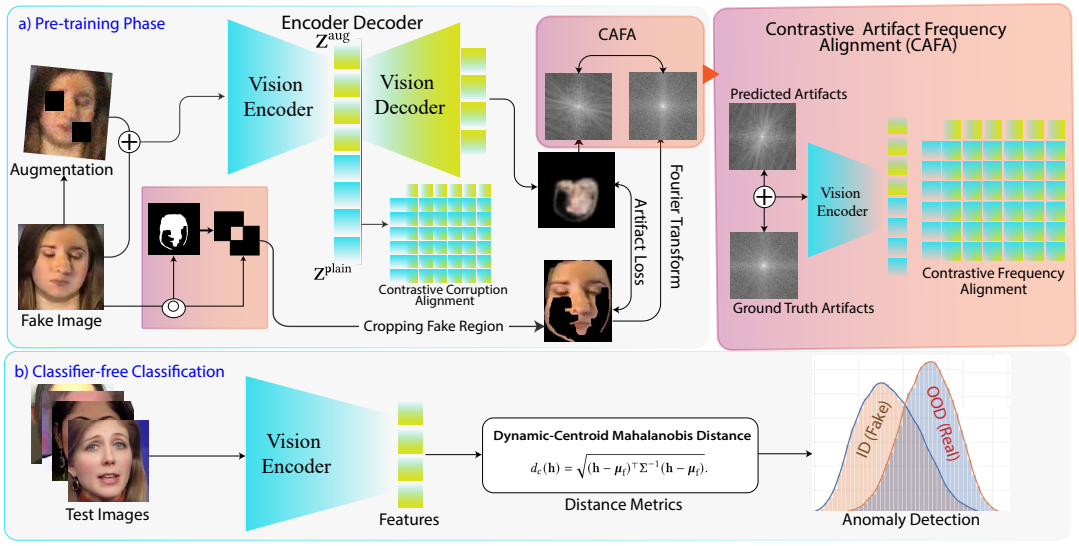


Figure 3: Overview of the FOCAL framework. Training Phase (top): Fake images and their augmentation are processed through a frozen CLIP ViT-L/14 encoder with LoRA-injected weights. The MAE decoder reconstructs the forgery region φ^{f} from patch embeddings, while spatial and frequency contrastive losses enforce robust, forgery-sensitive structure. Training uses only synthetic images. Detection Phase (bottom): The frozen encoder extracts features from a test image; Mahalanobis distance to the learned ID statistics (μ, Σ) determines whether the image is fake (in-distribution) or real (out-of-distribution anomaly).

3.3 Fake Blending Images Data Generation

Methods SBI [54] apply simple augmentations before blending the target image; however, this blending does not incorporate Generative Adversarial Network (GAN) artifacts, which we observe teach the network to blend boundaries rather than to learn forgery representations. Inspired by SBI, we propose the Fake Blending Images (FBI) dataset, which reconstructs a real image I_r using GFP-GAN [62], and CodeFormer [70] alternatively from FaceForensics++ real samples to form a fake image I_f. Face-restoration models do not introduce geometry inconsistencies or identity mismatches. We use it because GAN-based generators leave characteristic spectral and textural *fingerprints* [15, 69], such as over-smoothing of skin, reduced high-frequency detail, and periodic up-sampling artifacts, that are shared across many modern GAN-based manipulation pipelines. Following Self-Blended Images [54], which argues that *generic, hardly recognizable fake samples promote robust representations*, we replace SBI's augmented-real source with regeneration. This encourages the encoder to learn GAN-level fingerprints rather than blending artifacts, improving generalization to unseen manipulations (Sec. 4.2). FBI is not a forgery simulator but a label-free source of generic GAN traces. We extract facial landmarks from real images to generate random masks for forgery regions and blend them with fake images as:

$$I_{\text{FB}} = I_s \odot M + I_t \odot (1 - M), \tag{1}$$

where M denotes the blending mask generated from facial landmarks and $\odot$ represents element-wise multiplication. The mask M is constructed by computing the convex hull of the detected facial landmarks (which isolates the face from the background and ensures the model focuses only on the area where the forgery occurs) and then applying random augmentations to increase shape diversity. This mask defines the region in which the reconstructed (fake) content is inserted into the original image. To further enhance variability, we apply geometric transformations, including scaling, translation, and slight deformation, to both the mask and the reconstructed image prior to blending. This process introduces inconsistencies not only at the boundaries, but also within the facial region, allowing the model to learn more generalized forgery artifacts rather than overfitting to simple blending cues. After obtaining a facial mask M, we used LaMa [57] to further generate GAN-edited images from FF++ real samples, resulting in a diverse manipulation dataset comprising 508,969 regenerated blended images and 508,969 LaMa-generated images (1,017,938 images in total). The resulting Fake Blending Image dataset, shown in Fig. 4, contains both realistic GAN-induced artifacts and spatial blending inconsistencies, providing a more challenging and diverse training signal for forgery detection models.

3.4 Vision Encoder

We adopt the CLIP ViT-L/14 vision encoder [50] as the backbone encoder f_{enc}. Input images are resized to 224×224 and normalized with CLIP-standard statistics. The encoder produces two outputs: a pooled CLS token representation $\mathbf{h} \in \mathbb{R}^{1024}$ and a sequence of patch embeddings $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_{256}] \in \mathbb{R}^{256\times1024}$ (excluding the CLS token), where $256 = 16 \times 16$ corresponds to the 14×14-pixel patch grid at the 224×224 input resolution. The choice of CLIP ViT-L/14 is motivated by its strong visual representations and pre-training on large-scale datasets. However, unlike prior methods that adapt CLIP for binary real/fake classification [12, 20], we use CLIP solely

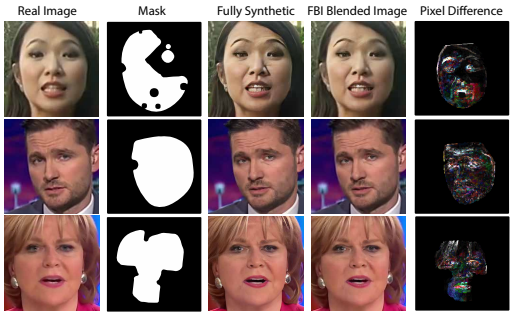


Figure 4: Illustration of some samples from the FBI dataset. The mask is used to blend a fully synthetic image and generate an FBI blended image. The pixel differences between the FBI-blended image and the real image are used to visualize the GAN artifacts.

as a feature extractor, with its representations shaped by forgery reconstruction.

3.5 Forgery Artifact Reconstruction

The core of FOCAL is a masked autoencoder (MAE) decoder [21] that reconstructs the forgery artifact map from the encoder's patch embeddings. Given patch features $\mathbf{E} \in \mathbb{R}^{256\times1024}$, the decoder f_{dec} maps them to a reconstructed artifact map:

$$\hat{\varphi}_i^{\text{f}} = f_{\text{dec}}(\mathbf{E}_i^{\text{f}}), \tag{2}$$

where $\hat{\varphi}_i^{\text{f}}$ is the predicted fake region and $\mathbf{E}_i^{\text{f}}$ is the feature embeddings of fake images. The decoder consists of a transformer with an input encoder dimension of 1024, projected to a decoder dimension of 512, followed by a pixel-shuffle upsampling head. We employ Artifact Reconstruction Loss (ARL) with pixel normalization for the reconstruction of fake regions as follows:

$$\mathcal{L}_{\text{ARL}} = \frac{1}{N}\sum_{i=1}^{N}\Bigg(\underbrace{\left\|(\hat{\varphi}_i^{\text{f}} - I_{\text{FB}})\odot M\right\|_2^2}_{\text{foreground forgery}\,\rightarrow\,\text{pixels}} + \underbrace{\left\|\hat{\varphi}_i^{\text{f}}\odot M^{\text{bg}}\right\|_2^2}_{\text{background}\,\rightarrow\,\text{black}}\Bigg) + \underbrace{\frac{1}{|\Omega|}\sum_{p\in\Omega}\left[-M_p\log\hat{M}_p - (1-M_p)\log(1-\hat{M}_p)\right]}_{\text{pixel-wise mask Cross Entropy}} \tag{3}$$

where M is the binary mask of manipulated pixels, $M^{\text{bg}} = 1 - M$ denotes the complementary background mask, $\hat{M}_p$ is the predicted mask probability at pixel p, and Ω is the set of all spatial pixels with $|\Omega|$ denoting its cardinality. The background loss penalizes responses in real/background regions, encouraging the decoder to predict black outputs outside manipulated areas without requiring explicit real-image targets. Pixel-wise mask cross-entropy supervises pixel-level mask prediction via binary cross-entropy, ensuring precise localization of manipulated regions. The reconstruction target is pixel-normalized as:

$$\bar{\varphi}_i^{\text{f}} = \frac{\varphi_i^{\text{f}} - \mu(\varphi_i^{\text{f}})}{\sqrt{\text{Var}(\varphi_i^{\text{f}}) + \epsilon}}, \tag{4}$$

where $\mu(\varphi_i^{\text{f}})$ and $\text{Var}(\varphi_i^{\text{f}})$ denote the mean and variance of the target artifact map, respectively.

Forgery Reconstruction Drives Forgery-Centric Features. The reconstruction objective in Equation (3) provides a direct inductive bias: the only way the encoder can minimize reconstruction loss is by encoding information about *where and how the image was manipulated*. Global facial identity, pose, and illumination are irrelevant to predicting fake region φ_i^{f}. They appear identically in both x_i^r and x_i^{f} and cancel in the residual. This design explicitly filters out identity and semantic information from the learned representation.

3.6 Augmentation-Aware Contrastive Learning

While reconstruction ensures forgery sensitivity, it does not explicitly encourage invariance to input perturbations present in OOD, which may preserve the forgery content. To produce robust features, we introduce an augmentation-aware contrastive objective. For each fake image x_i^{f}, we construct two views: a *plain* view x_i^{plain} and an *augmented* view x_i^{aug} (with standard data augmentations such as random blur, random crop, and noise applied). Both views are processed by the shared encoder f_{enc} to obtain pooled features $\mathbf{h}_i^{\text{plain}}$ and $\mathbf{h}_i^{\text{aug}}$, which are then projected through a lightweight embedding head:

$$\mathbf{z} = g(\mathbf{h}), \tag{5}$$

where $g : \mathbb{R}^{1024} \rightarrow \mathbb{S}^{255}$ maps features to the unit hypersphere.

The spatial contrastive loss follows the symmetric InfoNCE formulation [7, 60]:

$$\mathcal{L}_{\text{con}} = \frac{1}{2}\left[\ell_{\text{NCE}}(\mathbf{Z}^{\text{plain}}, \mathbf{Z}^{\text{aug}}) + \ell_{\text{NCE}}(\mathbf{Z}^{\text{aug}}, \mathbf{Z}^{\text{plain}})\right], \tag{6}$$

where $\mathbf{Z}^{\text{plain}}, \mathbf{Z}^{\text{aug}} \in \mathbb{R}^{B\times256}$ are batch matrices of projected embeddings, and:

$$\ell_{\text{NCE}}(\mathbf{Z}^a, \mathbf{Z}^b) = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(\mathbf{z}_i^a\cdot\mathbf{z}_i^b/\tau)}{\sum_{j=1}^{B}\exp(\mathbf{z}_i^a\cdot\mathbf{z}_j^b/\tau)}, \tag{7}$$

with temperature $\tau = 0.1$. The positive pair $(\mathbf{z}_i^{\text{plain}}, \mathbf{z}_i^{\text{aug}})$ corresponds to two views of the same fake image; all other batch elements serve as negatives. This objective encourages the encoder to produce features that are invariant to augmentation-induced perturbations while preserving inter-sample discriminability with respect to forgery content.

3.7 Frequency-Domain Contrastive View

Manipulation artifacts often leave distinctive traces in the frequency domain [3, 40, 49]—for instance, GAN-generated faces exhibit periodic spectral patterns, and blending operations suppress high-frequency content at manipulation boundaries. To explicitly encourage the encoder to capture these spectral cues, we introduce a frequency-domain contrastive view.

Given a fake image x_i^{f}, we construct a high-frequency emphasized view $\tilde{x}_i^{\text{freq}}$ as follows. First, we compute the 2D Fast Fourier Transform (FFT) and apply a radial high-pass filter as follows:

$$X_{\text{HP}} = \mathcal{F}^{-1}\left[\mathcal{F}[x_i^{\text{f}}]\odot M_{\text{HP}}\right], \tag{8}$$

where $\mathcal{F}$ denotes the 2D FFT with frequency shift and M_{HP} is a binary mask that suppresses spatial frequencies below a cutoff

radius $r_c = \rho \cdot \min(H, W)$ with $\rho = 0.08$. The high-pass output X_{HP} is min-max normalized per channel and blended with the original image as follows:

$$\tilde{x}_i^{\text{freq}} = \alpha_{\text{f}} \cdot \hat{X}_{\text{HP}} + (1 - \alpha_{\text{f}}) \cdot x_i^{\text{f}}, \tag{9}$$

with mixing coefficient $\alpha_{\text{f}} = 0.7$. This view emphasizes high-frequency manipulation artifacts while retaining spatial context.

The frequency-domain contrastive loss mirrors Equation (6):

$$\mathcal{L}_{\text{CFA}} = \frac{1}{2}\left[\ell_{\text{NCE}}(\mathbf{Z}^{\text{plain}}, \mathbf{Z}^{\text{freq}}) + \ell_{\text{NCE}}(\mathbf{Z}^{\text{freq}}, \mathbf{Z}^{\text{plain}})\right], \tag{10}$$

where $\mathbf{Z}^{\text{freq}}$ contains the projected embeddings of the frequency views extracted using a CLIP-based encoder. By contrasting the plain view with the frequency-enhanced view, we encourage the encoder to align representations of spatial and spectral forgery, yielding features that capture evidence of manipulation across both domains.

3.8 Overall Objective Loss

The total training objective combines reconstruction and contrastive losses:

$$\mathcal{L} = \lambda_r \mathcal{L}_{\text{ARL}} + \lambda_c \mathcal{L}_{\text{con}} + \lambda_{\text{f}} \mathcal{L}_{\text{CFA}}, \tag{11}$$

where $\lambda_r = 0.6$, $\lambda_c = 0.2$, and $\lambda_{\mathbf{f}} = 0.2$. Each term serves a distinct role: $\mathcal{L}_{\text{ARL}}$ ensures forgery-centric encoding by requiring the features to support artifact map reconstruction, $\mathcal{L}_{\text{con}}$ promotes augmentation-invariant but forgery-discriminative structure, and $\mathcal{L}_{\text{CFA}}$ injects spectral sensitivity into the representation.

3.9 OOD Detection

Out-of-distribution (OOD) detection is formulated as a distance score function, $d(\cdot)$, such as Guo et al. [18] on features extracted through a frozen encoder f_{enc} trained in a previous stage. The distance function assigns higher scores to ID data (fake images) and lower scores to OOD data (real images). The input can be classified as follows:

$$\mathcal{D}_{\text{test}}(x_i, d, \alpha) = \begin{cases} x_i \in ID & \text{if } d(x_i) > \alpha \\ x_i \in OOD & \text{if } d(x_i) < \alpha \end{cases} \tag{12}$$

This OOD formulation is a natural fit for the one-class setting, as it requires no classifier training and generalizes to unseen forgery types by leveraging the geometry of the learned feature space. We do not claim a new distance metric; instead, as a design choice for our scoring function, we evaluated several standard OOD distance formulations on our learned features, including Euclidean distance, k-NN distance, Isolation Forest scoring, and per-class Mahalanobis distance. Following common practice in the Mahalanobis-based OOD literature [18, 36], we adopt DCMD as it empirically gave the largest margin between OOD and ID distributions on our representation. Given the **training set** features $\mathbf{h}_i$, we estimate the fake centroid [36] $\boldsymbol{\mu}_{\text{f}}$ and shared covariance $\boldsymbol{\Sigma}$ via the Ledoit-Wolf shrinkage estimator [34] which provides a robust covariance estimate for stable Mahalanobis distance computation in high-dimensional feature spaces. The Mahalanobis distance for a test sample $\mathbf{h}$ is:

$$d_c(\mathbf{h}) = \sqrt{(\mathbf{h} - \boldsymbol{\mu}_{\text{f}})^\top \Sigma^{-1} (\mathbf{h} - \boldsymbol{\mu}_{\text{f}})}. \tag{13}$$

OOD (real) images tend to lie farther from the learned fake-artifact distribution, yielding lower scores, while ID (fake) images remain closer to the centroid and yield higher scores. This formulation requires no learned classifier or test-time labels and directly evaluates the quality of the learned representation geometry; the central contribution of FOCAL lies in the representation that makes this geometry separable, not in the distance metric itself.

4 Experiments

4.1 Implementation Details

To validate the FOCAL framework, we evaluate its cross-dataset generalization and robustness by pre-training on controlled synthetic data and testing on diverse real-world benchmarks. We aim to demonstrate that forgery-centric representations offer a more stable foundation for detection than traditional face-centric models. The following sections detail our data preparation, architectural configuration, and the OOD-based evaluation protocols used to assess our learned feature geometry.

4.1.1 ***Datasets.*** We pre-train FOCAL on the proposed FBI dataset generated from real samples of FaceForensics++ (FF++) [51] training set, which provides masks and fake images for the extraction of forged regions. For cross-dataset evaluation, we test on six standard benchmarks: CelebDF-v2 (CDF-v2) [39], DFDC [13], DFDCp [13], DFD [51], WildDeepFake (WDF) [72], and FFIW [71]. These datasets exhibit significant domain shifts across forgery type, target identity, background, and compression quality.

4.1.2 ***Training Details.*** We use the CLIP ViT-L/14 vision encoder as the backbone and fine-tune with Parameter-Efficient Fine-tuning (PEFT), freezing the encoder and injecting LoRA weights into the QKV layers. All input images are aligned, cropped to the face region using MediaPipe [42], and resized to 224×224. The MAE decoder has an encoder dimension of 1024, a decoder dimension of 512, and 256 input patches. Training uses AdamW [41] with learning rate 1×10^{-4}, weight decay 0.05, and batch size 32 for 15 epochs. Contrastive temperature is $\tau = 0.1$. The embedding head projects 1024-dim features through $1024 \rightarrow 512 \rightarrow 256$ with GELU activation and ℓ_2 normalization. Training is conducted on 2 NVIDIA A100 GPUs.

4.1.3 ***Evaluation Protocols.*** We evaluate the method on an OOD detection task by extracting features from a frozen encoder and computing distances using the Dynamic-Centroid Mahalanobis Distance (DCMD), thereby directly assessing the representation geometry without a classifier. Using AUC for comparison against classifier-based baselines, we further test on DF40 across the FF++ domain [29] (see Table 4). Our approach outperforms prior methods by 3.5% on FF++, despite being trained only on fake samples from the proposed FBI dataset, while others use binary classifiers. This indicates that forgery-centric features generalize better across datasets, whereas methods trained on both real and fake data can develop biases that hinder generalization.

4.2 Cross-Dataset Generalization

Table 1 reports cross-dataset generalization, where prior methods are trained and fine-tuned with binary classifiers, while FOCAL is trained on the FBI and uses classifier-free detection via DCMD. FOCAL achieves the highest average AUC of 90.4%, outperforming the previous best (FSFM) by 4.6%. This supports our hypothesis that

Table 1: Benchmark comparison on cross-dataset deepfake detection (AUC). Prior methods are supervised binary classifiers trained on FaceForensics++ (FF++) and evaluated under their original protocols without target-domain fine-tuning. In contrast, FOCAL is trained only on fake samples from the proposed FBI dataset and performs classifier-free detection using DCMD scoring (Section 3.9) on frozen encoder features. Since training data and inference protocols differ across methods, this table serves as a benchmark reference comparison. A protocol-matched evaluation under identical FBI pretraining and DCMD settings is provided in Table 2.

Method	Venue	CDF-v2	DFDC	DFDCp	DFD	WDF	FFIW	Avg.
F3Net [49]	ECCV'20	0.789	0.718	0.749	0.844	0.728	0.649	0.746
SPSL [40]	CVPR'21	0.799	0.724	0.770	0.871	0.702	0.794	0.777
Face X-Ray [38]	CVPR'20	0.794	0.692	0.738	0.872	0.694	0.651	0.740
SBI [54]	CVPR'22	0.868	0.717	0.848	0.827	0.703	0.866	0.805
RECCE [4]	CVPR'22	0.823	0.696	0.734	0.891	0.756	0.711	0.769
UCF [66]	ICCV'23	0.837	0.742	0.770	0.867	0.774	0.697	0.781
CORE [45]	CVPRW'22	0.809	0.721	0.720	0.882	0.724	0.710	0.761
DCL [6]	AAAI'22	0.882	0.750	0.769	0.921	0.776	0.863	0.827
LSDA [64]	CVPR'24	0.898	0.735	0.812	0.956	0.756	0.879	0.839
DELOCATE [24]	IJCAI'24	0.824	0.745	0.853	**0.963**	0.834	0.793	0.835
FSFM [61]	CVPR'25	0.861	0.759	0.930	0.933	0.828	0.835	0.858
Effort [65]	ICML'25	0.897	0.824	0.901	0.791	0.814	0.839	0.843
SD^2 [27]	ICLR'26	–	0.803	–	–	0.828	–	0.815
FOCAL (**ours**)	–	**0.921**	**0.834**	**0.951**	0.941	**0.854**	**0.924**	**0.904**

Table 2: Representation quality comparison using the proposed Mahalanobis distance (AUC %). All methods are pretrained on the FBI dataset and evaluated without fine-tuning or classifier training. Encoder features are scored via DCMD, isolating representation quality from downstream classifiers.

Method	DFDC		CDF-v2		WDF		AVG.	
	AUC	P@avg	AUC	P@avg	AUC	P@avg	AUC	P@avg
DELOCATE [24]	0.574	0.352	0.684	0.394	0.544	0.317	0.601	0.354
FSFM [61]	0.609	0.468	0.706	0.543	0.625	0.429	0.647	0.480
FOCAL (**ours**)	**0.833**	**0.706**	**0.921**	**0.783**	**0.854**	**0.649**	**0.869**	**0.713**
Δ vs. FSFM	+22.4	+23.8	+21.4	+24.0	+22.8	+22.0	+22.2	+23.3
Δ vs. DELOCATE	+25.9	+35.4	+23.6	+38.9	+30.9	+33.2	+26.8	+35.9

forgery-centric training yields features better suited for unseen manipulations than face-centric approaches. The gains over FSFM and DELOCATE are notable: both learn real-face representations via self-supervised objectives, introducing biases toward identity and facial semantics that limit cross-dataset transfer. In contrast, FOCAL learns manipulation-driven features that capture shared artifacts, such as blending boundaries, frequency inconsistencies, and texture discontinuities, thereby enabling stronger generalization.

4.3 Representation Quality Analysis

Table 2 is the main apple-to-apple comparison to evaluate representation quality without fine-tuning or classifiers. All methods are pre-trained on the FBI dataset, and encoder features are evaluated on target datasets using DCMD with 50 PCA dimensions and the Ledoit-Wolf covariance. FOCAL surpasses FSFM and DELOCATE by +22.2% and +26.8% in average AUC. It also achieves a P@avg of 0.713 for$k = \{5, 10, 20, 30, 50\}$, exceeding FSFM and DELOCATE by +23.3 and +35.9, indicating that forgery-centric features produce more separable distributions under DCMD. These gains stem from stronger representations rather than from downstream classifiers. Table 6 further shows the advantage of FBI over Self-Blending Images (SBI) [54]. SBI blends augmented versions of the same real image, encouraging detection of blending boundaries rather than forgery artifacts. In contrast, the FBI reconstructs images with facial regeneration methods before blending, injecting realistic GAN artifacts alongside spatial inconsistencies (Section 3.3). When all methods are pretrained on SBI and evaluated with DCMD, performance drops significantly, confirming that FBI's richer forgery signals are essential for learning transferable representations. This highlights that pretraining data with genuine manipulation artifacts, rather than simple blending cues, ensures generalization of FOCAL.

Table 3: Ablation study on loss components. Each row removes one objective from the full model.

$\mathcal{L}_{\text{rec}}$	$\mathcal{L}_{\text{con}}$	$\mathcal{L}_{\text{freq}}$	DFDC	CDF-v2
✓			0.766	0.850
	✓		0.712	0.791
✓	✓		0.805	0.893
✓		✓	0.791	0.877
	✓	✓	0.744	0.829
✓	✓	✓	**0.834**	**0.921**

4.4 Ablation Studies

4.4.1 ***Loss Component Analysis.*** Table 3 examines the contribution of each loss component. The reconstruction loss $\mathcal{L}_{\text{rec}}$ alone

Table 4: Cross-manipulation evaluations within the FF++ domain of DF40 [29]. The proposed model is trained on the proposed FBI dataset, while other models are trained on FF++ multiple forgeries, tested on rest. The baseline results are sourced from [68].

Detector	UniFace	BlendFace	MobileSwap	e4s	FaceDancer	FSGAN	Inswap	SimSwap	Avg.
RECCE [5]	0.898	0.832	0.925	0.683	0.848	0.949	0.848	0.768	0.844
SBI [54]	0.724	0.891	0.952	0.750	0.594	0.803	0.712	0.701	0.766
UCF [67]	0.831	0.827	0.950	0.731	0.862	0.937	0.809	0.647	0.824
AltFreezing [63]	0.947	0.951	0.851	0.605	0.836	0.958	0.927	0.797	0.859
AuthGuard [53]	0.955	-	-	0.896	0.856	0.953	0.950	0.857	0.916
FOCAL (Ours)	**0.967**	**0.959**	**0.976**	**0.932**	**0.928**	**0.964**	**0.978**	**0.901**	**0.951**

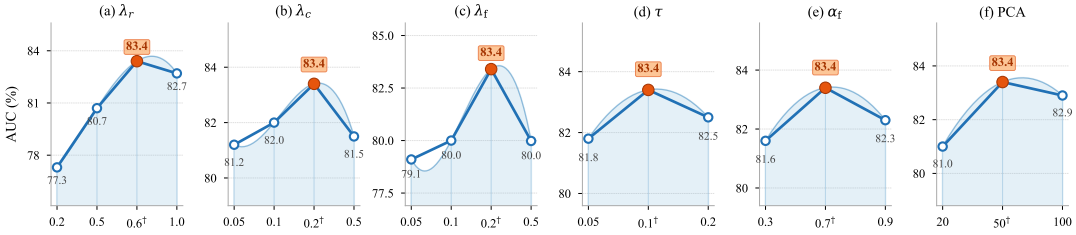


Figure 5: Ablation study on key hyperparameters. Default values (†) are used in the full model. Evaluated via DCMD AUC (%) on DFDC.

achieves 76.6% on DFDC using the Mahalanobis distance, confirming that artifact reconstruction is a strong representation-learning objective. Adding the spatial contrastive loss ($\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{con}}$) improves this to 80.5%, demonstrating that augmentation invariance strengthens the representation geometry. The frequency contrastive loss further boosts the full model to 83.4%, contributing 2.9% over the two-loss variant. Notably, contrastive learning alone ($\mathcal{L}_{\text{con}}$ only, 71.2%) performs substantially worse than reconstruction alone, confirming that the forgery reconstruction objective is the primary driver of representation quality.

Table 5: Ablation on encoder backbone. All variants use the full FOCAL training pipeline. Evaluated via Mahalanobis distance AUC (%) on DFDC and CDF-v2.

Backbone	Feat. Dim	DFDC	CDF-v2
ResNet-50 [22]	2048	69.7	81.9
ViT-B/16 [14]	768	74.3	86.0
DINOv2-L/14 [46]	1024	77.9	88.0
CLIP ViT-B/16 [50]	768	78.6	89.3
CLIP ViT-L/14 [50]	1024	**83.4**	**92.1**

4.4.2 ***Hyperparameter Sensitivity.*** Fig. 5 examines sensitivity to key hyperparameters. The model is moderately sensitive to the reconstruction weight λ_r (77.3% at $\lambda_r = 0.2$ vs. 83.4% at $\lambda_r = 0.6$), reinforcing that the reconstruction objective is the dominant training signal. Contrastive temperature τ and frequency parameters show less sensitivity, with performance remaining above 81% across all tested values. PCA dimension 50 provides the best balance between denoising and information preservation for DCMD scoring.

Table 6: Evaluation of the Self-Blending Images (SBI) dataset with all methods assessed without fine-tuning or classifier training. Encoder features are scored via DCMD, enabling direct comparison with FBI-based results and isolating representation quality across datasets.

Method	DFDC		CDF-v2		WDF		AVG.	
	AUC	P@avg	AUC	P@avg	AUC	P@avg	AUC	P@avg
DELOCATE [24]	0.532	0.317	0.626	0.346	0.508	0.286	0.555	0.316
FSFM [61]	0.555	0.425	0.641	0.504	0.577	0.388	0.591	0.439
FOCAL (**ours**)	**0.764**	**0.648**	**0.832**	**0.711**	**0.781**	**0.593**	**0.792**	**0.651**
Δ vs. FSFM	+20.9	+22.3	+19.1	+20.7	+20.4	+20.5	+20.1	+21.2
Δ vs. DELOCATE	+23.2	+33.1	+20.6	+36.5	+27.3	+30.7	+23.7	+33.5

4.4.3 ***Backbone Analysis.*** Table 5 compares different encoder backbones under the full FOCAL training pipeline. CLIP ViT-L/14 achieves the best performance across both evaluation protocols. The improvement from ViT-B/16 to ViT-L/14 (74.3% → 83.4% on DFDC) suggests that larger encoder capacity enables more expressive forgery representations. DINOv2-L/14 performs comparably to CLIP ViT-B/16, indicating that the pre-training data distribution (CLIP's web-scale image-text pairs vs. DINOv2's curated image collections) influences the quality of the forgery features learned through our pipeline.

4.4.4 ***Evaluation of Robustness against Common Perturbations.*** We conduct a robustness analysis on the FF++ dataset against six common environmental perturbations, as illustrated in Fig. 6. We measure the detection AUC across five levels of increasing severity for corruptions, including Gaussian Blur, JPEG Compression, and Gaussian Noise. While existing state-of-the-art methods, including CLIP, FSFM, and EFFORT, experience significant performance degradation as image quality decreases, FOCAL

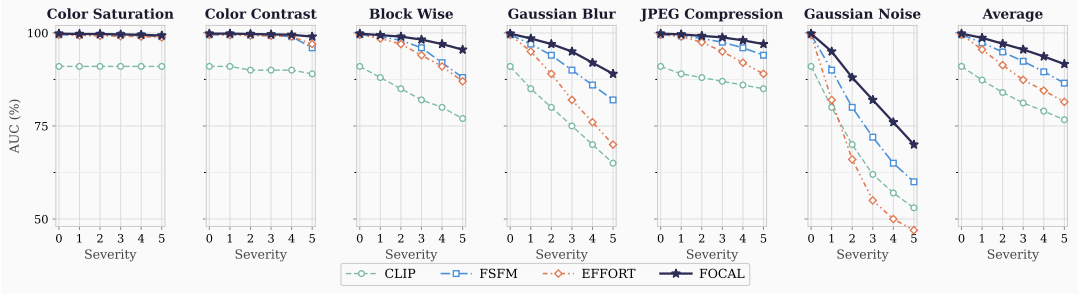


Figure 6: Robustness analysis on FF++. We evaluate AUC under six common perturbations, such as Gaussian Blur and JPEG compression, across multiple severity levels. FOCAL exhibits lower performance degradation compared to SOTA.

demonstrates remarkable resilience across all categories. Notably, in high-severity scenarios (Level 5), FOCAL maintains a superior AUC, particularly against frequency-destructive corruptions like JPEG compression and Gaussian blur. This stability is attributed to our augmentation-aware contrastive learning and frequency-domain alignment (CFA), which forces the encoder to prioritize robust forgery-centric features over fragile, quality-dependent pixel cues, ensuring reliable detection in unconstrained real-world environments.

4.4.5 ***Unsupervised Distance-based Classification.*** To evaluate the impact of the distance-based classification mechanism, we conduct an ablation study comparing various unsupervised anomaly detection algorithms for Out-of-Distribution (OOD) outlier-based classification on the DFDC dataset, as illustrated in Fig. 8. Each plot shows the density distribution of In-Distribution (ID → Fake) and Out-of-Distribution (OOD → Real) samples as a function of the calculated distance. Our results indicate that traditional metrics, such as **Euclidean Distance** (AUC = 56.18%) and classic anomaly detectors like **Isolation Forest** (AUC = 57.93%), struggle to find a separable boundary in the high-dimensional embedding space. While the standard **Mahalanobis Distance** provides a baseline for separation, it suffers from significant distribution overlap. In contrast, the **DCMD** achieves the highest performance with an (AUC of 83.41%), effectively maximizing the margin between real and fake distributions. By treating deepfake detection as an OOD task on the DFDC dataset, DCMD effectively maximizes the distance between the learned forgery cluster and the unseen authentic distribution, confirming that our framework's "fake-only" pre-training creates a distinct, outlier-sensitive manifold.

4.5 Forgery Localization

4.5.1 ***Qualitative Forgery Analysis.*** Fig. 7 further illustrates the qualitative evaluation of FOCAL across four distinct forgery types: UniFace, SimSwap, FSGAN, and BlendFace. For each forgery method, FOCAL produces masks that approximate the ground-truth artifact regions, accurately describing the swapped facial boundaries and blending artifacts. In contrast, competing methods such as HiFINet and SIDA often produce coarse, over-segmented masks that fuse background regions with manipulation evidence. DELOCATE, while producing more structured outputs than the former

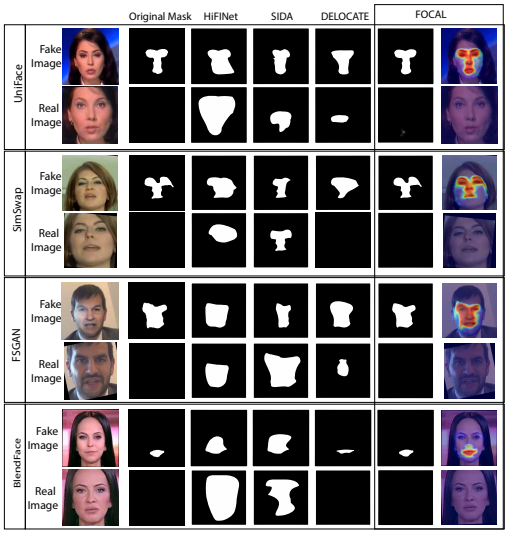


Figure 7: Qualitative comparison of real masks and fake masks predicted from existing methods and FOCAL. All models are trained on the FBI dataset, including both real and fake samples, except the proposed approach, which is trained only on fake samples.

two, still exhibits notable boundary imprecision and fails to suppress false positives in homogeneous skin regions, as observed in the Real Image rows of the SimSwap and BlendFace categories. A particularly noteworthy observation is FOCAL behavior on real images: unlike competing methods that produce spurious activations even for unmanipulated faces, our reconstructed artifact maps yield near-zero responses on authentic samples. This behavior is a direct consequence of the proposed $\mathcal{L}_{AFL}$ loss function, which conditions the model to activate exclusively in the presence of forgery-specific distributional deviations, rather than responding to generic facial features. This distinction is particularly critical in practical deployment scenarios, where false-positive rates on real images can severely undermine system credibility.

4.5.2 ***Quantitative Forgery Analysis.*** The reconstructed artifact map $\hat{\rho}^f$, as defined in Eq. 2, serves as a direct spatial indicator of detected manipulation evidence. As shown in Table 7, our method

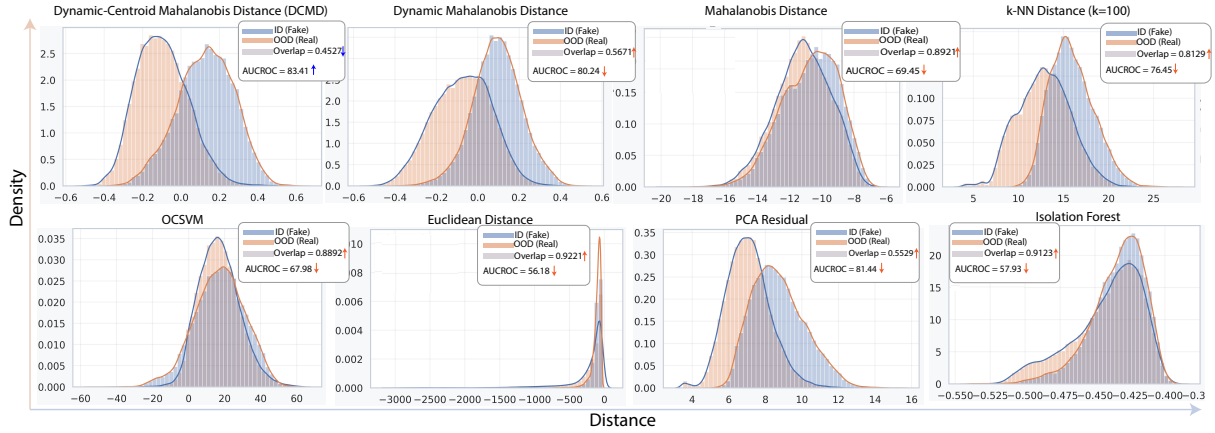


Figure 8: Ablation study on different distance functions for OOD outlier-based classification using the DFDC test set.

consistently outperforms specialized localization models across all key metrics. As evidenced by the significant AUC (0.921) and IoU (0.81) gains on CelebDF-V2 GT masks obtained using a similar technique as DELOCATE [24], human annotations are not feasible, and expert users cannot spot subtle forgeries on the massive CelebDF-V2 benchmark. Our proposed forgery-centric training captures the spatial extent of manipulations more accurately than methods that rely on traditional face-centric representations. Furthermore, using $\mathcal{L}_{AFL}$ effectively minimizes pixel-level discrepancies and yields cleaner and more reliable localization maps, as indicated by the lowest MAE of 0.024 and better IoU, Dice, and pixel-level F1 scores. This suggests that our one-class setting provides a stronger inductive bias for localization than binary classification approaches.

Table 7: Quantitative Localization performance comparison with SOTA localization methods on CDF-v2. Metrics are computed at the pixel level between predicted masks and ground-truth artifact maps.

Method	IoU↑	Dice↑	Pixel-F1↑	AUC↑	MAE↓
TruFor [17]	0.44	0.61	0.59	0.68	0.082
HiFINet [19]	0.45	0.62	0.60	0.69	0.079
SIDA [25]	0.46	0.64	0.62	0.70	0.076
DELOCATE [24]	0.77	0.78	0.68	0.82	0.031
FOCAL (Ours)	**0.81**	**0.79**	**0.72**	**0.921**	**0.024**

5 Limitations and Future Work

FOCAL is trained on the proposed FBI dataset, which primarily captures GAN-induced fingerprints from two distinct GAN-based generators. While the cross-dataset evaluation in Table 1 already demonstrates strong performance, we expect further gains from incorporating a broader range of generators. We adopted blended regeneration and LaMa specifically because they support face regeneration and inpainting, which, in turn, enable the automatic extraction of reliable ground-truth masks; most other deepfake generators do not offer this capability. Since ground-truth masks are unavailable for manipulation methods such as FaceSwap and FaceShifter, we introduced the FBI dataset to circumvent the need for manual pixel-level annotation, which is infeasible for visually imperceptible deepfakes.

6 Conclusion

We present FOCAL, a forgery-centric one-class representation learning framework for deepfake detection. Unlike prior face-centric methods that learn representations by modeling real face structure or identity, FOCAL directly learns and encodes manipulation artifacts through an encoder-decoder architecture trained to reconstruct forged regions. Combined with augmentation-aware spatial- and frequency-domain contrastive objectives, this design yields features explicitly grounded in evidence of forgery rather than in facial semantics. Framing deepfake detection as an OOD problem, FOCAL enables classifier-free detection using DCMD in the learned feature space. Our experiments provide evidence for a key insight: representations learned by modeling *what forgeries look like* transfer and perform more effectively across datasets than representations learned by modeling *what real faces look like*. Without any fine-tuning or classifier training, FOCAL achieves 83.3% AUC on DFDC via DCMD, outperforming FSFM and DELOCATE under identical conditions.

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GenAI Usage Disclosure

We used GenAI tools to assist with equation corrections and GPT-5 for PyTorch boilerplate code. AI usage was limited to improving presentation clarity and grammar checks.

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